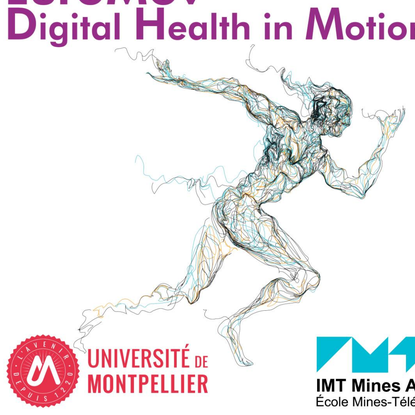
EuroMov Digital Health in Motion



DEEP LEARNING FOR PHYSICAL LOAD ESTIMATION: INSIGHTS FROM THE VILOAD VIDEO DATASET

Nadi a Lahya, EuroMov Digital Health in Motion, Univ Montpellier, IMT Mines Ales, Ales, France
Nouha TALEB-SALAH, EuroMov Digital Health in Motion, Univ Montpellier, IMT Mines Ales, Ales, France
Hajer Fradi, EuroMov Digital Health in Motion, Univ Montpellier, IMT Mines Ales, Ales, France
Oussama Ben-Ammar, SyCoIA, IMT Mines Ales, Ales, France
Pierre Slangen, EuroMov Digital Health in Motion, Univ Montpellier, IMT Mines Ales, Ales, France
Jacky Montmain, SyCoIA, IMT Mines Ales, Ales, France

1 Background and motivation
Why This Work?

MSDs are among the leading causes of sick leave in industrial sectors, driven by awkward postures, repetitive motions, and manual load handling.

The REBA method integrates load weight directly into ergonomic risk scoring. Yet existing automated pipelines focus on posture alone and ignore the load being handled, a critical gap for real-world MSD prevention.

Wearable IMUs capture load-related dynamics but are costly, intrusive, and poorly scalable in industrial settings.

Goal: Estimate handled load (empty / light / heavy) from RGB video alone, using deep learning, no wearable sensors required.

2 Key contribution
What we bring

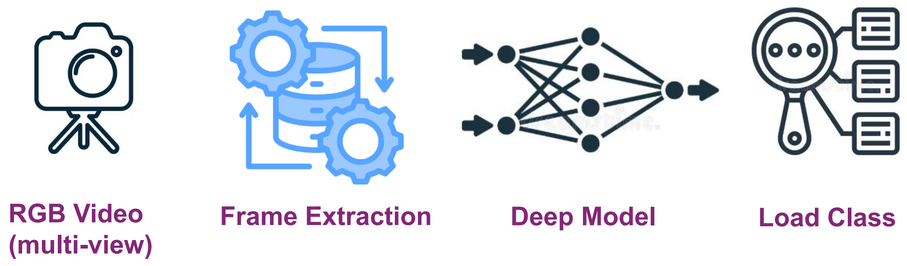
Contribution

Contribution 1
First investigation of video-based load estimation using deep learning for ergonomic assessment

Contribution 2
Two complementary architectures: MobileNetV2(frame-level) and R3D-18(video-level spatio-temporal)

Contribution 3
Introduction ofViLoad— a new multi-view RGB + IMU dataset for load estimation research, openly shared with the community

3 Methodology
Processing Pipeline



0 KG Empty · 5 KG Light · 10-15 KG Heavy

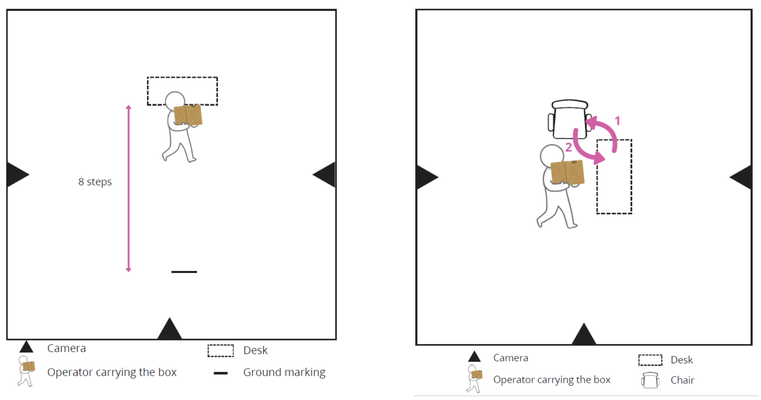
Heavy load threshold: 10 kg (women) / 15 kg (men), accounts for physiological differences.

4 Experimental setup
ViLoad Dataset

8 healthy volunteers (3F / 5M, avg. age 21.5) · No prior MSDs · Equipped with Xsens Awinda IMUs

Recorded simultaneously with 3 synchronized cameras (4K, 60 fps): static Xsens reference + 2 smartphones

144 Video · 3 Viewpoints · 2 Tasks



Split: 70% train / 15% val / 15% test ·
1,440 frames extracted (10 per video) for image-level model

5 Deep Learning Architectures

MobileNetV2
Frame-level CNN
Transfer learning from ImageNet
Frozen feature extractor + custom head:
AvgPool → BN → FC(256) × 2 → Softmax(3)

R3D-18
3D ResNet
Spatio-temporal convolutions Captures motion dynamics across frames
Fine-tuned final FC → 3 classes

Both trained with AdamW (lr = 1e-4) on Tesla T4 GPU
Data augmentation for MobileNetV2 (rotation, flip, zoom, brightness)

5 Results

RESULTS – FRAME-LEVEL (MOBILENETV2
IMAGE CLASSIFICATION PERFORMANCE

51% Accuracy) · 0.48 Recall · 0.45 F1-score · 0.83 Empty recall

TRUE \ PRED	EMPTY	LIGHT	HEAVY
Empty	0.83	0.04	0.13
Light	0.47	0.18	0.34
Heavy	0.43	0.18	0.39

Empty class well identified · ⚠ Significant confusion between **Light & Heavy,** similar visual appearance limits discrimination

RESULTS – VIDEO-LEVEL (R3D-18)
SPATIO-TEMPORAL CLASSIFICATION

48% Accuracy · 0.75 Empty recall

CLASS	PRECISION	RECALL	F1
Empty	0.400	**0.750**	0.522
Light	**1.000**	0.400	0.571
Heavy	0.250	0.200	0.222

Light class: perfect precision (1.00) but low recall , predictions correct when made, but many missed · Heavy: hardest to discriminate visually

Limitations & Discussion

Small dataset (144 videos · 8 operators) limits generalization

Visually similar sequences across load levels - same postures, angles, lighting

Only 10 frames/video extracted for image model, risk of overfitting to specific details

IMU modality collected but not yet exploited, multimodal fusion reserved for future work

6 Conclusion & Perspectives
Key Takeaways & Next Steps

- ✓ First exploration of vision-only deep learning for physical load estimation in an ergonomics context , MobileNetV2 and R3D-18 both show promising initial results (~50% accuracy on a small dataset).
- ✓ The ViLoad dataset is a novel, structured contribution, 144 videos across 3 load levels, 2 task types, 3 viewpoints, with paired IMU data, filling a clear gap in available resources.
- ✓ Temporal modeling (R3D-18) does not consistently outperform frame-level (MobileNetV2), suggesting the dataset size currently limits the advantage of spatio-temporal features.
- ✓ Future work: dataset expansion (more participants, varied environments) + multimodal fusion (video + IMU) + integration with pose-based REBA scoring for fully automated ergonomic risk assessment.

23/04/2026

References
[1] 2019. Adaptable Workstations for Human–Robot Collaboration: A ReconfigurableFramework for Improving Worker Ergonomics and Productivity. PP (March 2019),1–1
[2] Valentin Bazarevsky, Ivan Grishchenko, Karthik Raveendran, Tyler Zhu, FanZhang, and Matthias Grundmann. 2020. BlazePose: On-device Real-time BodyPose tracking.

Contact
Email: nouha.taleb-salah@mines-ales.fr
Date: 23/04/2026

Acknowledgements
- This work is part of the SANTÉ'OPTIM research project, co-funded by the Région Occitanie, France.
- Experiments conducted on the AIHM platform (Ales Imaging and Human Metrology) · IMT Mines Alès.

- CC BY-NC-ND 4.0 · 8 participants · Informed consent obtained.